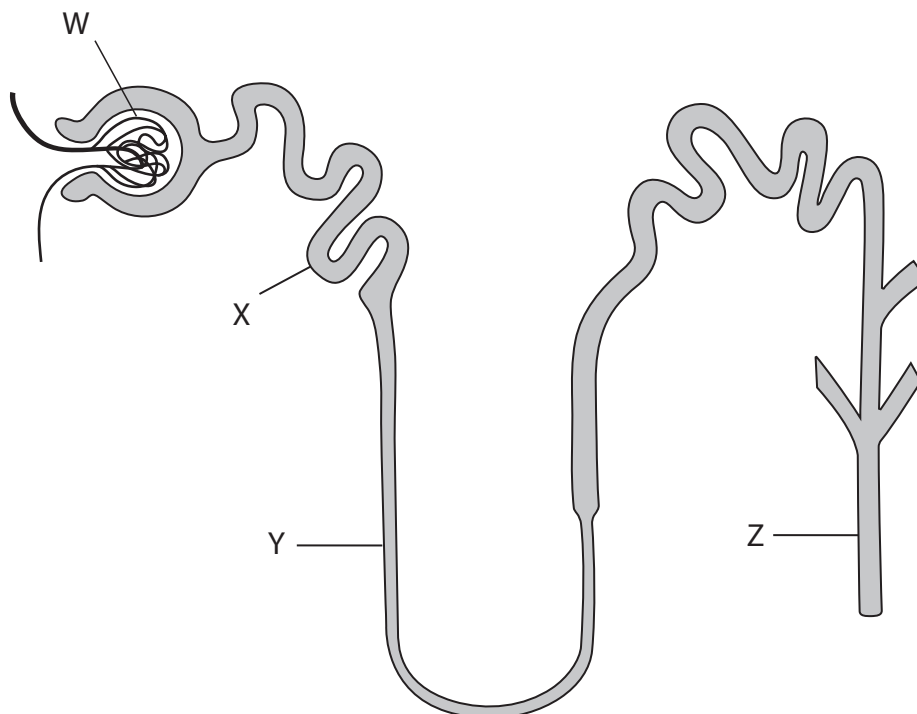


2 The kidney is an organ involved in the excretion of waste materials and in the regulation of blood volume and plasma concentration.

(a) The functional unit of the kidney is the nephron.

The diagram shows a single nephron.



(i) In which part of the kidney are structures W and X located?

(1)

- ☐ **A** cortex
- ☐ **B** inner medulla
- ☐ **C** outer medulla
- ☐ **D** renal pelvis

(ii) In which structure does ultrafiltration take place?

(1)

- ☐ **A** W
- ☐ **B** X
- ☐ **C** Y
- ☐ **D** Z

(iii) Which structures form part of the loop of Henle?

(1)

- ☐ **A** X only
- ☐ **B** X and Y
- ☐ **C** Y only
- ☐ **D** Y and Z

(b) Urea is a toxic substance that is excreted from the body.

Explain how ultrafiltration removes urea from the blood.

(2)

- (c) The kidney controls blood volume and plasma concentration by regulating the loss of water in urine.

Some mammals have adaptations to their kidneys that enable them to live in different habitats.

The table shows information about two small mammals.

Mammal	Urine concentration / mOsmol dm^{-3}	Relative number of mitochondria in the thick ascending loop of Henle	Sodium ion concentration of extracellular fluid in the medulla / mmol dm^{-3}
brown rat	2900	3.6	45
kangaroo rat	5500	5.5	95

Deduce why the kangaroo rat is more successful than the brown rat at living in desert habitats.

(3)

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(Total for Question 2 = 8 marks)